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TRENDS IN MORTALITY AMONG INDIVIDUALS WITH TYPE 1 DIABETES FROM 15 TO 25 YEARS POST-DIAGNOSIS IN DIFFERENT ERAS: A CROSS-SECTIONAL, REAL-WORLD OBSERVATIONAL STUDY

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Background and aims: Improvements in blood lipid control, glucose management and cardiovascular treatment have reduced macrovascular morbidity and mortality in people living with type 1 diabetes (T1D). This study examined whether mortality differs among people living with T1D in Belgium between 15 and 25 years post-diagnosis diagnosed between 1985 and 1998 (group A), compared to those diagnosed between 1998 and 2009 (group B).

Materials and methods: Data were extracted from the Initiative for Quality Improvement and Epidemiology in Diabetes. Individuals with a clinical diagnosis of T1D and diagnosed before the age of 30, who were 15 years-post diagnosis and had a known vital status, were selected. The total study population was divided into two study groups according to the calendar year of diabetes onset, with each group covering 10 read-out years. Group A included 345 individuals diagnosed between 1985-1998 (read-out years 2001-2011); group B included 948 individuals diagnosed between 1998-2009 (read-out years 2012-2022). Both groups were followed until 25 years after diagnosis for their vital status. Kaplan-Meier methods assessed cumulative mortality. Cox regression determined associations between patient characteristics at baseline and risk of mortality.

Results: At 15 years post-diagnosis, mean HbA1c was higher in group A (67.4 vs. 64.8 mmol/mol (8.3 vs. 8.1 %), $p < 0.05$), while obesity was more common in group B (11.7% vs. 17.6% $p < 0.05$). Mean LDL cholesterol and systolic blood pressure did not differ between the two groups. Antihypertensive therapy use was greater in group A (15.7% vs 11.3%, $p < 0.01$), while lipid-lowering therapy was more common in group B (10.4% vs. 15.2% $p < 0.05$). At 25 years post-diagnosis, the cumulative incidence of mortality was higher in group A (5.5% [95% CI 3.4-8.3] versus 2.1% [95% CI 1.2-3.5] $p < 0.05$). The association between patient characteristics at 15 years post-diagnosis and the mortality rate is shown in table 1.

Conclusion: Survival rates were better for those diagnosed in a later era (1998-2009), independently of age at diabetes onset, sex, and other baseline patient characteristics. A higher HbA_{1c} level 15 years post-diagnosis and higher age at diagnosis were positively correlated with an increased risk for mortality 25 years post-diagnosis.

Table 1: Association between patient characteristics at baseline and risk of mortality – 10 year follow up

	Mortality	
<i>Individuals</i>	<i>N = 1070</i>	
<i>Events</i>	<i>n = 31</i>	
	HR (95% CI)	p-value *
Study group B (diabetes onset 1998 – 2009)	0.324 (0.135-0.777)	0.0116
Age at diabetes onset, years	1.086 (1.026-1.149)	0.0044
Sex, women	0.411 (0.168-1.006)	0.0514
Characteristics at 15 years post-diagnosis		
HbA _{1c} (%)	1.498 (1.165-1.926)	0.0016
LDL cholesterol (mmol/L)	0.643 (0.372-1.113)	0.1145
SBP (mm Hg)	1.005 (0.974-1.037)	0.7472
BMI (kg/m ²)	0.878 (0.759-1.014)	0.0773

* Reference: Study group A (diabetes onset from 1985 – 1998), Men